Teacher Answer Key

Mechanical Oscillations – Ex 3: Analysis of Graphs

1. Frequency of the Oscillator

From graph (a), the elongation x completes a full cycle (period T) between $t = 0$ and $t = 0.4$ s.

Reading from the axis: $\boxed{T = 0.4 \text{ s}}$

The frequency f is the inverse of the period:

$$f = \frac{1}{T} = \frac{1}{0.4} = 2.5$$

Frequency: $\boxed{f = 2.5 \text{ Hz}}$

(Natural angular frequency: $\omega = 2\pi f = 5\pi \text{ rad/s}$)

2. Identification of Energy Graphs

We observe the initial state at $t = 0$ from Graph (a):

The curve starts at $x = 0$. Thus, at $t = 0$, the Elastic Potential Energy is:

$$EPE = \frac{1}{2}Kx^2 = 0$$

Since Mechanical Energy is conserved ($ME = KE + EPE = \text{const}$), if $EPE = 0$, then KE must be maximum.

Matching:

- **Graph (B)** starts at 0. $\rightarrow$ **Elastic Potential Energy (EPE)**.
- **Graph (A)** starts at maximum (0.8 J). $\rightarrow$ **Kinetic Energy (KE)**.

3. Equations and Constants

a) Equation of $x(t)$ and Max Speed:

The x vs t line graph from figure (a) is sinusoidal with a constant amplitude X_m and period T_0. Thus, the solution is of the form:

$$x = X_m \sin(\omega t + \varphi)$$

Determination of Parameters:

- $X_m = 4 \text{ cm} = 0.04 \text{ m}$ (from Graph a).
- $\omega = 5\pi \text{ rad/s}$.

Determination of Phase Angle (φ):

At $t = 0$, the graph shows $x(0) = 0$.

$$X_m \sin(\varphi) = 0 \implies \sin(\varphi) = 0$$

This gives two possible values: $\varphi = 0$ rad or $\varphi = \pi$ rad.

To filter, we check the initial velocity (v_0) from the slope of the graph at $t = 0$. The curve is going upwards, so $v_0 > 0$.

$$v = x' = X_m \omega \cos(\omega t + \varphi)$$

At $t = 0$: $v_0 = X_m \omega \cos(\varphi) > 0$. Since X_m and ω are positive, $\cos(\varphi)$ must be positive.

- If $\varphi = 0, \cos(0) = 1 > 0$ (Accepted).
- If $\varphi = \pi, \cos(\pi) = -1 < 0$ (Rejected).

Thus, $\boxed{\varphi = 0}$.

Equation: $\boxed{x = 0.04 \sin(5\pi t)}$ (SI)

Maximum Speed (V_{max}):

$$V_{max} = X_m \omega = 0.04(5\pi) = 0.2\pi \approx 0.628 \text{ m/s}$$

Result: $\boxed{V_{max} \approx 0.63 \text{ m/s}}$

b) Calculate M and K:

From Graph (b), the Mechanical Energy is constant at $\boxed{ME = 0.8 \text{ J}}$.

Calculate Stiffness K:

$$ME = \frac{1}{2} K X_m^2$$

$$0.8 = \frac{1}{2} K (0.04)^2 \implies 1.6 = K(0.0016)$$

$$K = \frac{1.6}{0.0016} = 1000$$

Result: $\boxed{K = 1000 \text{ N/m}}$

Calculate Mass M:

Using $\omega^2 = \frac{K}{M}$:

$$(5\pi)^2 = \frac{1000}{M} \implies 25\pi^2 = \frac{1000}{M}$$

Assume $\pi^2 \approx 10$: $250 = \frac{1000}{M}$

$$M = \frac{1000}{250} = 4$$

Result: $\boxed{M = 4 \text{ kg}}$

c) Verification at $t = 0.24$ s:

Using Equations:

$$x(0.24) = 0.04 \sin(5\pi \cdot 0.24) = 0.04 \sin(1.2\pi)$$

$\sin(1.2\pi) = \sin(216^\circ) \approx -0.588$

$$x \approx -0.0235 \text{ m} = -2.35 \text{ cm}$$

$$v(0.24) = x' = 0.2\pi \cos(1.2\pi)$$

$\cos(1.2\pi) \approx -0.809$

$$v \approx 0.628(-0.809) \approx -0.51 \text{ m/s}$$

Using Graphs:

- **Graph (a):** At $t = 0.24$s (slightly past $T/2 = 0.2$), the curve is negative and going down. -2.35 cm is consistent.
- **Graph (b):** At $t = 0.24$s (0.04s after crossing 0), Energy is rising.
  $EPE = \frac{1}{2}(1000)(-0.0235)^2 \approx 0.28$ J.
  $KE = \frac{1}{2}(4)(-0.51)^2 \approx 0.52$ J.
  $Sum = 0.8$ J. (Matches ME).

4. Period Relation

The period of oscillation is $T = 0.4$ s.

Looking at the energy graphs (A) or (B), the energy completes a cycle (e.g., from max to max) every 0.2 s.

Relation:

$$\boxed{T_{osc} = 2 \times T_{energy}}$$

> *Energy oscillates twice as fast as the elongation because energy depends on squared terms (x^2 or v^2), which are positive in both positive and negative half-cycles of motion.*